Zomatao DataSet Analysis

Exploratory Data Analysis & Buisness Insight Report

Prepared By	Data Science Intern – Krish Urkude
Dataset	Zomato Restaurant Dataset (Kaggle)
Records Analysed	31,257 restaurants (post-cleaning)
Scope	Bengaluru Restaurant Ecosystem
Tools	Python · Pandas · Seaborn · Matplotlib · WordCloud

1. Executive Summary

This report presents a comprehensive analysis of the Zomato restaurant dataset covering 31,257 restaurants across Bengaluru. The analysis spans data cleaning, exploratory data analysis, eight visualisations, and five strategic recommendations to inform Alfido Tech's partnership, content, and platform decisions.

Metric	Value	Metric	Value
Total Restaurants	31,257	Avg Rating	3.71 / 5
Unique Locations	516	Avg Cost (2 pax)	Rs. 599
Unique Cuisines	552	Online Order %	64.7%
Top Cuisine	North Indian	Table Booking %	15.9%

2. Dataset Description

Source: Kaggle – Bhanu Pratap Biswas. 56,252 raw rows, 13 columns. All records represent restaurants listed in Bengaluru.

Column	Description	Cleaning Applied
name	Restaurant name	Deduplicated
rate	Rating string e.g. 4.1/5	Regex-extracted to float
votes	User vote count	Cast to numeric
location	Neighbourhood area	Null rows removed
cuisines	Multi-valued cuisine list	Primary cuisine extracted
approx_cost(for two)	Price for 2 people	Numeric; 99th pct cap
online_order	Accepts online orders?	Standardised to Yes/No
book_table	Table booking available?	Standardised to Yes/No
dish_liked	Customer favourite dishes	Used for word cloud

3. Data Cleaning

- Removed duplicates and malformed rows: 56,252 raw -> 31,257 clean records retained.
- Extracted numeric rating from 'X/5' strings; discarded non-numeric entries ('Opening Soon', 'Top Floor').
- Extracted numeric cost; capped extreme outliers at 99th percentile (Rs. 3,000).
- Parsed primary cuisine (first listed) from comma-separated multi-value field.
- Standardised online_order and book_table fields to binary Yes/No; removed review-text bleed-over.
- Retained partial-null rows for dish_liked; applied dropna only during word cloud generation.

4. Exploratory Data Analysis & Visualisations

4.1 Rating Distribution

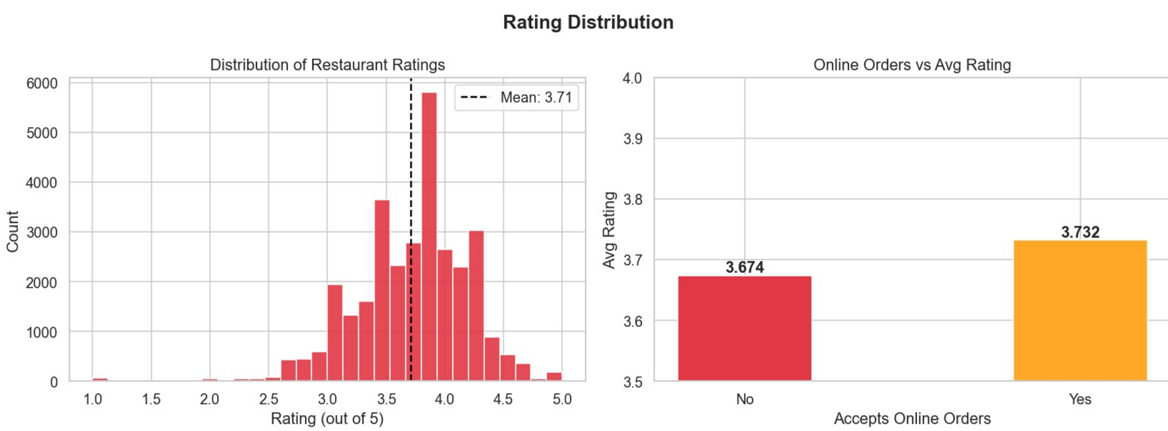

Figure 1 – Rating Histogram and Online Order Impact on Rating

Ratings cluster between 3.4 and 4.1 with a mean of 3.71. The distribution is slightly left-skewed, indicating rare poor ratings that pull the mean down. Restaurants with online ordering score 3.73 vs 3.67 for offline-only – digital adoption correlates with higher customer satisfaction.

4.2 Location vs. Dining Type Heatmap

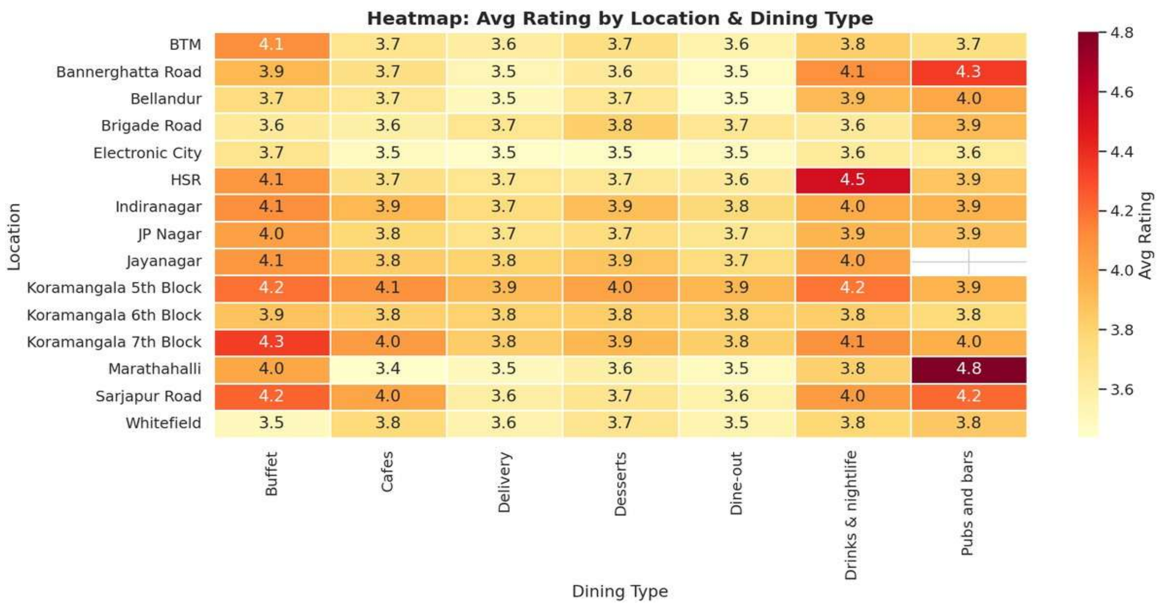

Figure 2 – Average Rating by Location and Dining Category

Premium neighbourhoods such as Indiranagar, Koramangala, and Whitefield consistently achieve ratings of 3.9–4.2. BTM, despite having the most restaurants, shows moderate average ratings – a classic saturation effect. Dessert shops and bar segments excel across high-income zones.

4.3 Top Cuisines by Restaurant Count

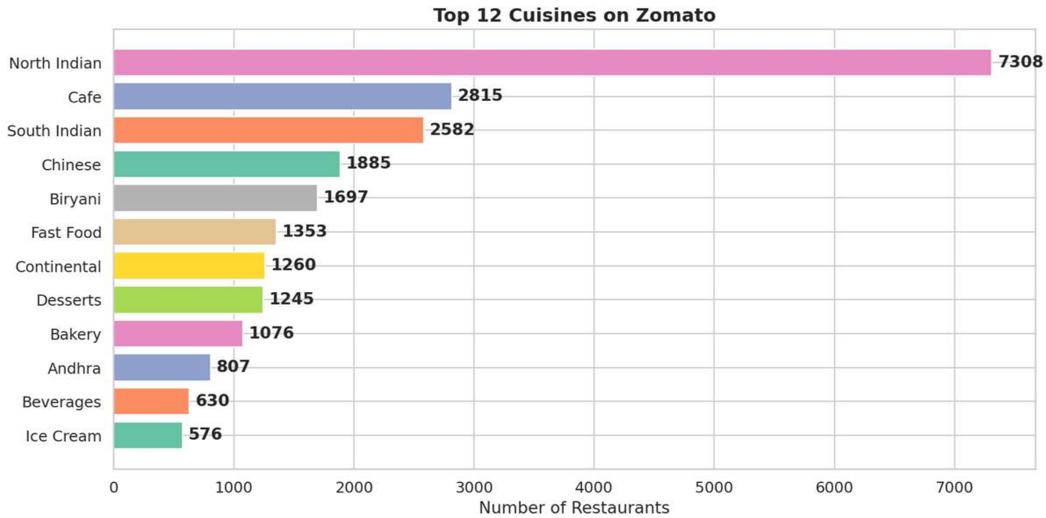

Figure 3 – Top 12 Cuisines on Zomato Bengaluru

North Indian leads with 7,308 restaurants – nearly 3x the next (Cafe: 2,815). Chinese, South Indian and Biryani complete the top 5. The remaining 547 cuisines form a long tail of niche opportunities with low competition and high engagement potential.

4.4 Price vs Rating

Figure 4 – Approx Cost for Two vs Rating (Scatter + Trend Line)

A moderate positive correlation (r = 0.367) exists between price and rating. Restaurants priced Rs. 1,500–3,000 tend to achieve ratings above 4.0. Budget eateries under Rs. 300 cluster near 3.5, but select high-rating budget spots represent 'hidden gem' content.

4.5 Location Hotspots

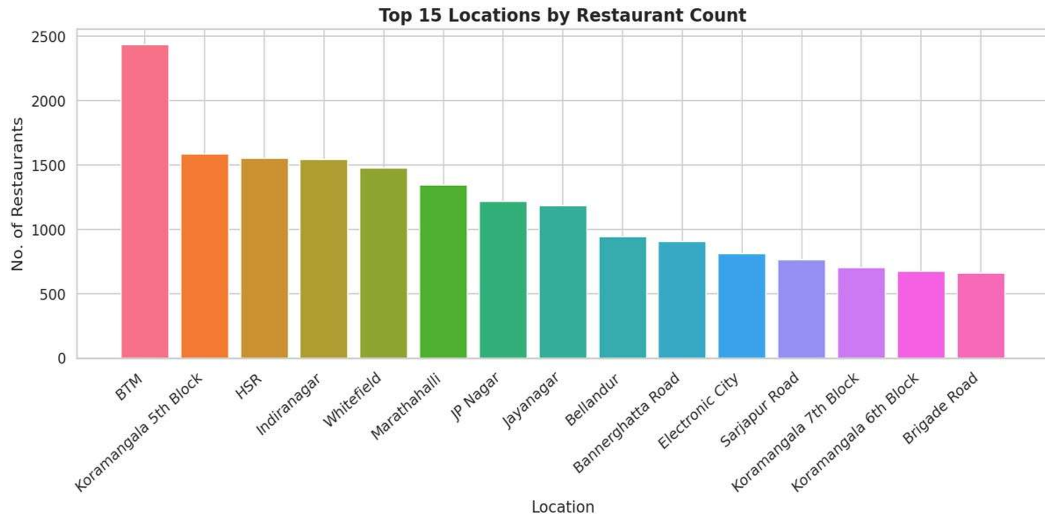

Figure 5 – Top 15 Locations by Restaurant Count

BTM (2,434), Koramangala 5th Block (1,587), and HSR (1,553) are the dominant hubs. The top five locations together hold approximately 35% of all restaurants, making them priority targets for geo-targeted campaigns and neighbourhood editorial content.

4.6 Cuisine vs Average Rating

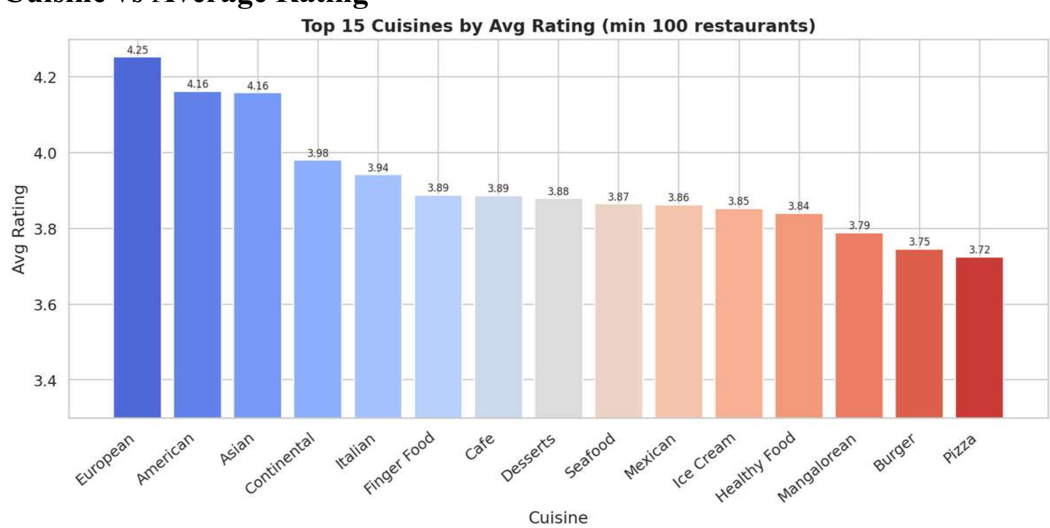

Figure 6 – Top 15 Cuisines by Average Rating (min 100 restaurants)

European (4.25), American (4.16), and Asian (4.16) top the quality chart despite lower volume. North Indian does not rank in the top quality tiers a strategic content angle: speciality cuisines are underserved yet command premium user satisfaction.

4.7 Word Cloud - Most Liked Dishes

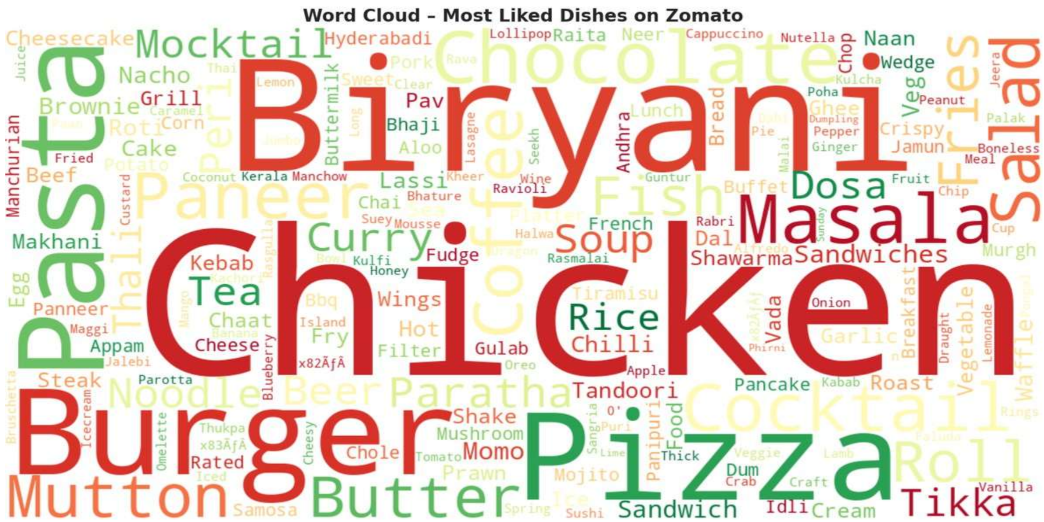

Figure 7 – Word Cloud of Dishes Most Liked by Customers

Biryani, Butter Chicken, Masala Dosa, and Momos are the most frequently cited favourites. Pasta, Waffles, and Cold Coffee reflect growing global-food preferences. These dish signals directly power a personalised recommendation engine.

4.8 Feature Availability vs Rating

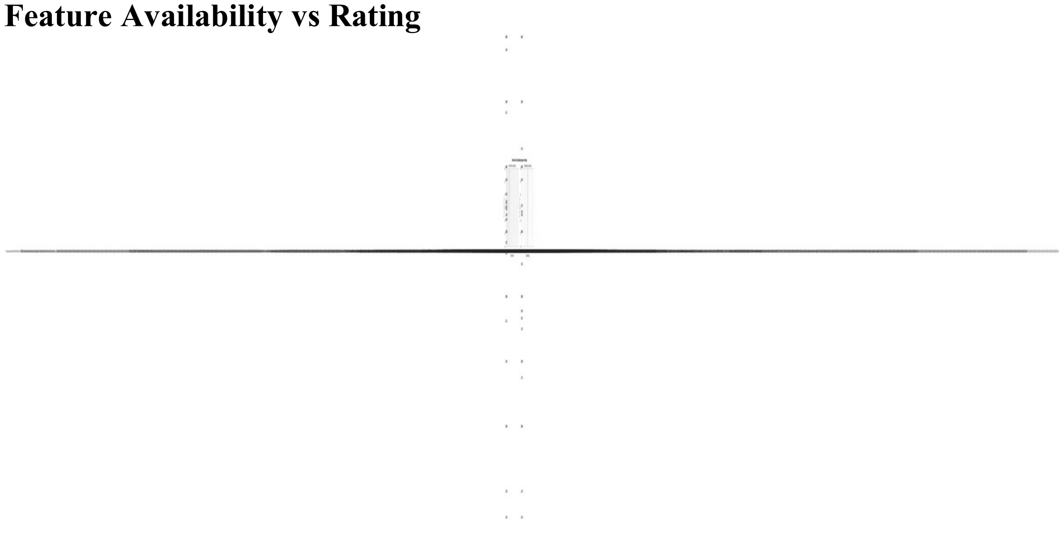

Figure 8 – Online Order and Table Booking Impact on Average Rating

Online ordering: 3.73 (Yes) vs 3.67 (No). Table booking shows a stronger lift: 3.92 (Yes) vs 3.70 (No) – a 0.22-point gap. Restaurants offering table booking deliver a measurably more premium experience, consistently earning higher customer ratings.

5. Key Findings

- Ratings are clustered (3.4–4.1); outlier high-rated restaurants are the most valuable discovery assets.
- BTM, Koramangala, and Indiranagar are the definitive hotspots – 35%+ of all listings.
- North Indian is most common but not highest-rated; European/American/Asian cuisines outperform on quality.
- Price-rating correlation (r=0.367): premium dining reliably earns better ratings; budget gems also exist.
- Table booking strongly linked to higher ratings (+0.22 pts); only 15.9% of restaurants offer it.
- Online order at 64.7% leaves 35% still offline – a clear digitisation growth opportunity.
- Biryani, Butter Chicken, Masala Dosa top dish preferences – dish-level personalisation is feasible.

6. Business Recommendations for Alfido Tech

Rec 1 – Partner with High-Rating Speciality Cuisines

European, American, and Asian restaurants achieve ratings of 4.15+ yet are underrepresented in volume. Alfido Tech should recruit these as premium partners with curated 'Editor's Choice' badges and priority placement in search and home-feed results.

Expected Impact: Premium revenue tier, improved platform NPS, competitive differentiation from mass-market listings.

Rec 2 – Geo-Targeted Hyper-Local Campaigns

BTM, Koramangala 5th Block, HSR, Indiranagar, and Whitefield host 35%+ of all restaurants. Geo-fenced push notifications, 'Best of [Neighbourhood]' weekly guides, and dedicated area landing pages will maximise daily active usage and advertising revenue from these commercial zones.

Expected Impact: Higher CTR, strong local retention, and premium ad packages for Bengaluru's top commercial areas.

Rec 3 – 'Hidden Gems' Editorial Programme

Cross-referencing price and rating surfaces restaurants priced under Rs.400 with ratings above 4.0. A weekly 'Hidden Gem' feature in-app and in newsletters drives budget-conscious discovery and generates highly shareable social content at near-zero production cost.

Expected Impact: User acquisition among 18–28 age group, organic social virality, brand trust and editorial authority.

Rec 4 – Table Booking Onboarding Initiative

With only 15.9% offering table bookings yet a 0.22-point rating advantage, this is a clear feature gap. Alfido Tech should offer onboarding support, waive early commissions, and surface a prominent 'Book a Table' CTA to convert intent-browsers into committed diners.

Expected Impact: Increased GMV per visit, higher restaurant retention rates, and a premium user experience signal.

Rec 5 – Dish-Level Personalisation Engine

The dish_liked dataset reveals consistent fan-favourite dishes citywide. Building a dish-preference profile per user and surfacing restaurant suggestions by dish match-score – rather than generic cuisine filters – enables deeper personalisation and higher conversion.

Expected Impact: Higher order conversion, longer sessions, and a meaningful competitive advantage over cuisine-only filters.

7. Conclusion

This analysis of 31,257 Bengaluru restaurants on Zomato reveals a dynamic, geographically concentrated ecosystem. Key quality drivers are table booking, online order adoption, premium pricing, and speciality cuisine focus. The five recommendations provide Alfido Tech with a clear, evidence-based roadmap to grow partnerships, user engagement, and gross merchandise value.

Future extensions could include: NLP sentiment analysis on review text, time-series rating trend modelling, predictive rating ML models (XGBoost/Random Forest), and restaurant growth-cohort analysis to identify rising stars proactively.

Report prepared as part of the Alfido Tech Data Science Internship Programme.

Dataset: Zomato by Bhanu Pratap Biswas - Kaggle.com | Tools: Python, Pandas, Seaborn, Matplotlib, WordCloud, ReportLab